PRE LAB 08
DEWAGEDARA D.M.E.S.
E/21/087

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- what is indexing?
  - An index is data structure that improves the speed of data retrieval on a table at the cost of additional write and storage. It is similar to the index of a book - it helps you quickly find what you are looking for.

- How indexing works,
  - when a query runs, DBMS checks whether it can use an index to avoid full table scans
  - indexes are usually created on columns that are frequently use in,
    1. where clauses
    2. JOIN clauses
    3. ORDER BY,
    4. GROUP BY
  - Most databases use B-Trees for indexing

- when to use indexing,
  1. You query large tables frequently
  2. You filter / search by specific column often
  3. You join tables on certain keys
  4. You sort results often

- Issues with indexing,
  1. slow down INSERT, UPDATE and DELETE.
  2. Takes extra storage space.
  3. over-indexing can degrade performance

- index stores,
  1. The indexed column value
  2. A reference to the actual row in the table
  3. often sorted for fast lookup

- Types of Indexing,
  1. primary key
  2. unique index
  3. composite index
  4. clustered index
  5. Non-clustered index

- ~~ana~~ Query optimization is the process of improving a query's efficiency by minimizing,
  1. CPU usage
  2. Disk I/O
  3. Memory usage
  4. Response time.

  It is handle by the query optimizer, a part of the DBMS.

- key concepts in MySQL Query optimizations
  1. Use indexes effectively
  2. Select only needed columns
  3. ~~Ao~~ Avoid subqueries if possible.

- common Query optimization techniques
  1. Ex PLAIN - use
  2. Use indexes
  3. Use JOIN efficiently
  4. Normalize table.

TASK 01

```
mysql> SELECT d.dept_name, COUNT(*) AS engineer_count
    -> FROM employees e
    -> JOIN titles t ON e.emp_no = t.emp_no
    -> JOIN dept_emp de ON e.emp_no = de.emp_no
    -> JOIN departments d ON de.dept_no = d.dept_no
    -> WHERE t.title = 'Engineer'
    -> GROUP BY d.dept_name;
+--------------------+----------------+
| dept_name          | engineer_count |
+--------------------+----------------+
| Production         |          47286 |
| Development        |          55336 |
| Quality Management |          13213 |
| Research           |           2855 |
| Customer Service   |           2252 |
+--------------------+----------------+
5 rows in set (2.57 sec)
```

TASK 02

```
mysql> SELECT DISTINCT e.emp_no, e.first_name, e.last_name
    -> FROM employees e
    -> JOIN dept_manager dm ON e.emp_no = dm.emp_no
    -> JOIN titles t ON e.emp_no = t.emp_no
    -> WHERE e.sex = 'F'
    ->   AND t.title = 'Senior Engineer';
Empty set (0.00 sec)
```

TASK 03

```
mysql>
mysql> SELECT d.dept_name, t.title, COUNT(*) AS employee_count
    -> FROM employees e
    -> JOIN salaries s ON e.emp_no = s.emp_no
    -> JOIN dept_emp de ON e.emp_no = de.emp_no
    -> JOIN departments d ON de.dept_no = d.dept_no
    -> JOIN titles t ON e.emp_no = t.emp_no
    -> WHERE s.salary > 115000
    -> GROUP BY d.dept_name, t.title;
```

```
    -> GROUP BY d.dept_name, t.title;
+--------------------+--------------------+----------------+
| dept_name          | title              | employee_count |
+--------------------+--------------------+----------------+
| Marketing          | Senior Staff       |           1061 |
| Marketing          | Staff              |            855 |
| Finance            | Senior Staff       |            768 |
| Finance            | Staff              |            650 |
| Human Resources    | Senior Staff       |             42 |
| Human Resources    | Staff              |             27 |
| Production         | Engineer           |            178 |
| Production         | Senior Engineer    |            202 |
| Production         | Technique Leader   |             32 |
| Production         | Assistant Engineer |             16 |
| Development        | Engineer           |            258 |
| Development        | Senior Engineer    |            277 |
| Development        | Assistant Engineer |             52 |
| Development        | Technique Leader   |             21 |
| Development        | Senior Staff       |              2 |
| Quality Management | Engineer           |             17 |
| Quality Management | Senior Engineer    |             17 |
| Quality Management | Assistant Engineer |              1 |
| Quality Management | Technique Leader   |              6 |
| Sales              | Staff              |           7106 |
| Sales              | Senior Staff       |           8627 |
| Research           | Senior Staff       |             54 |
| Research           | Staff              |             51 |
| Research           | Technique Leader   |              7 |
| Research           | Engineer           |             12 |
| Research           | Senior Engineer    |             17 |
| Research           | Assistant Engineer |              4 |
| Customer Service   | Senior Staff       |            447 |
| Customer Service   | Staff              |            356 |
| Customer Service   | Senior Engineer    |             16 |
| Customer Service   | Engineer           |             11 |
+--------------------+--------------------+----------------+
31 rows in set (7.40 sec)
```

TASK 04

```
mysql> SELECT
    ->     e.first_name,
    ->     e.last_name,
    ->     TIMESTAMPDIFF(YEAR, e.birth_date, CURDATE()) AS age,
    ->     TIMESTAMPDIFF(YEAR, e.hire_date, CURDATE()) AS years_of_service,
    ->     e.hire_date AS joined_date
    -> FROM employees e
    -> WHERE
    ->     TIMESTAMPDIFF(YEAR, e.birth_date, CURDATE()) > 50
    ->     AND TIMESTAMPDIFF(YEAR, e.hire_date, CURDATE()) > 10;
```

```
| Elzbieta      | Zucker          |  68 |              35 | 1990-05-18
| Ung           | Marquardt       |  68 |              39 | 1985-10-04
| Kwun          | Schiettecatte   |  72 |              32 | 1993-05-14
| Hatsukazu     | Perry           |  63 |              33 | 1991-09-02
| Evgueni       | McDermid        |  69 |              35 | 1990-02-08
| Marin         | Zhang           |  67 |              31 | 1994-01-07
| Bojan         | Gihr            |  64 |              34 | 1990-11-13
| Babette       | Eldridge        |  61 |              33 | 1992-06-18
| Arunachalam   | Falck           |  64 |              30 | 1994-11-28
| Tayeb         | Pusterhofer     |  61 |              38 | 1986-09-11
| Berthier      | Braunschweig    |  68 |              37 | 1987-10-31
| Laurentiu     | Koblitz         |  68 |              39 | 1986-04-27
| Lorien        | Verhaegen       |  61 |              40 | 1985-02-08
| Mariusz       | Serna           |  73 |              32 | 1993-02-03
| Xiaoshan      | Alpin           |  65 |              38 | 1987-03-19
| Tru           | Werthner        |  68 |              27 | 1997-12-22
+---------------+-----------------+-----+-----------------+------------
+^C -- query aborted

300024 rows in set (0.71 sec)
```

TASK 05

```
mysql> SELECT e.first_name, e.last_name
    -> FROM employees e
    -> JOIN dept_emp de ON e.emp_no = de.emp_no
    -> JOIN departments d ON de.dept_no = d.dept_no
    -> WHERE d.dept_name != 'Human Resources';
```

```
| Faiza          | Matzat          |
| Padma          | Dalphin         |
| Zhaofang       | Radivojevic     |
| Duro           | Brender         |
| Zhongwei       | Egerstedt       |
| Anestis        | Swan            |
| Gregory        | Mayerwieser     |
| Wilmer         | Plesums         |
| Hinrich        | Marzano         |
| Feiyu          | Gulla           |
| Lena           | Giaccio         |
| Jingling       | Lalonde         |
| Perry          | Turnbull        |
| Rosalie        | Gammage         |
| Jamaludin      | Thorelli        |
| Chenye         | Walstra         |
| Geoffrey       | Marciano        |
| Kristinn       | Naudin          |
| Qingxiang      | Gips            |
| Susuma         | Baalen          |
| Shaunak        | Schlumberger    |
| Elvis          | Adachi          |
| Frazer         | Szmurlo         |
| Aloke          | Rikino          |
| Niteen         | Bierbaum        |
| Sajjad         | Gist            |
| Berto          | Hitomi          |
| Vitaly         | Rouquie         |
| Frazer         | Pearson         |
| Hausi          | Katiyar         |
| Mooi           | Vilarrasa       |
| Freyja         | Danlos          |
| Chuanyi        | Schaad          |
| Sadegh         | Genther         |
| Achillea^      | | -- query aborted
 Lichtman        |
+----------------+-----------------+
313817 rows in set (1.18 sec)
```